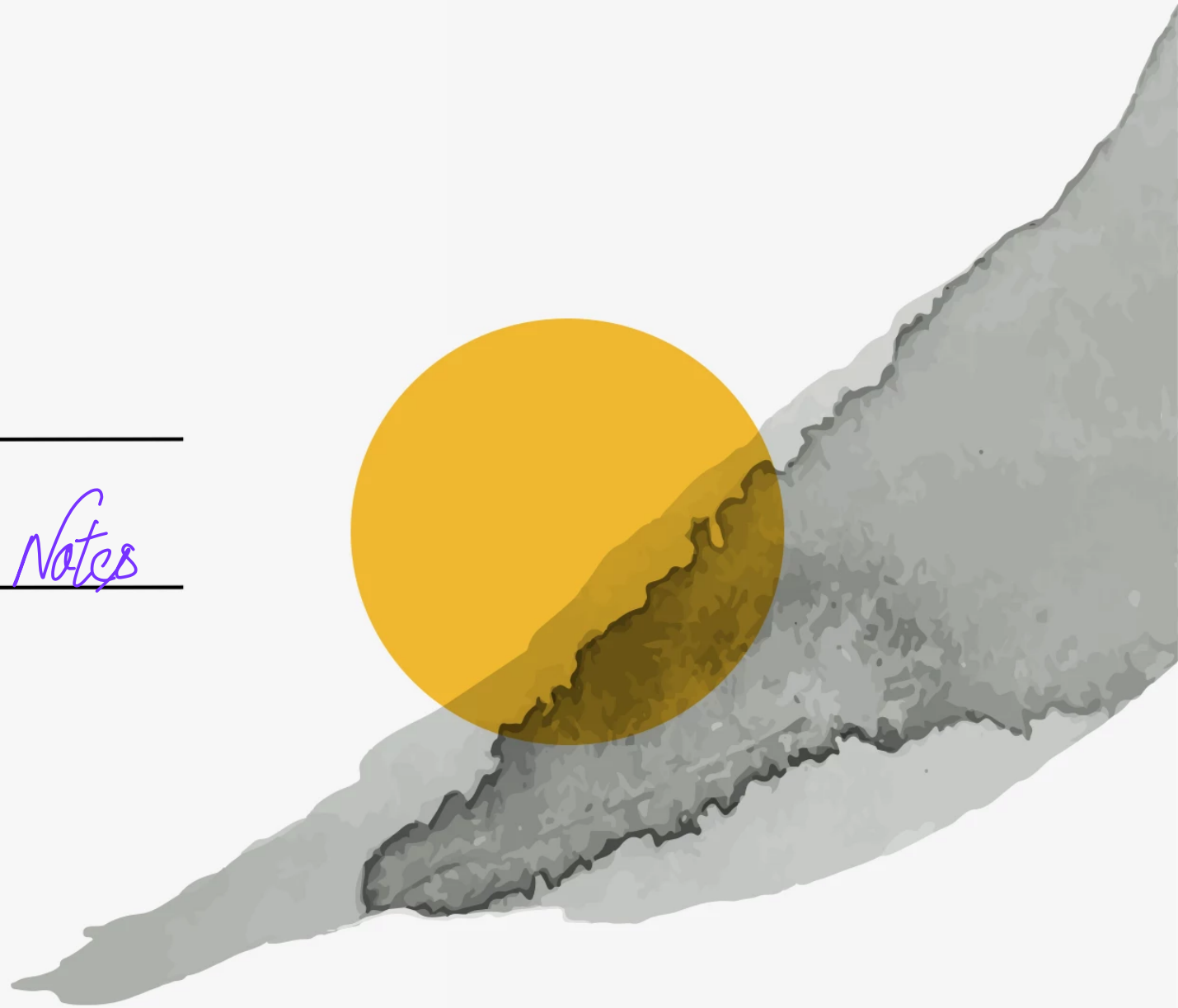
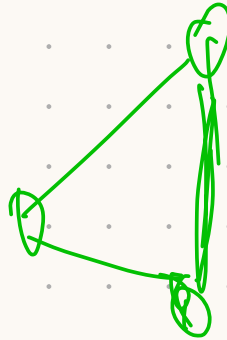


Notebook

Devashish Notes





3MC Math Epi course organisation

4 April 2022

Julien Arino   

Department of Mathematics & Data Science Nexus
University of Manitoba*

Canadian Centre for Disease Modelling

Canadian COVID-19 Mathematical Modelling Task Force

NSERC-PHAC EID Modelling Consortium (CANMOD, MfPH, OMNI/RÉUNIS)

Course objectives

Introduction to Mathematical Epidemiology

- Problems
- Methods

We will have these particular problems in mind:

1. Modelling techniques
2. Mathematical analysis of models
3. Computational analysis of models
4. Use of data

It is important to do the 4 interactively

About modelling

- Do not neglect this step
- Think outside the box
- Take the time
- Try to remain simple

Epidemiology

Definition

Epidemiology is the study and analysis of the distribution (who, when, and where), patterns and determinants of health and disease conditions in defined populations [[Wikipedia](#)]

Epidemiology is the study of how often diseases occur in different groups of people and why. Epidemiological information is used to plan and evaluate strategies to prevent illness and as a guide to the management of patients in whom disease has already developed [[BMJ](#)]

Trying to use it to study Info. Diffusion in Social N/W
then trying to decrease the time needed to
reach all Nodes by using Influence Maximization.

Outbreak

sudden increase in occurrences of a disease in a particular time and place.

Endemic

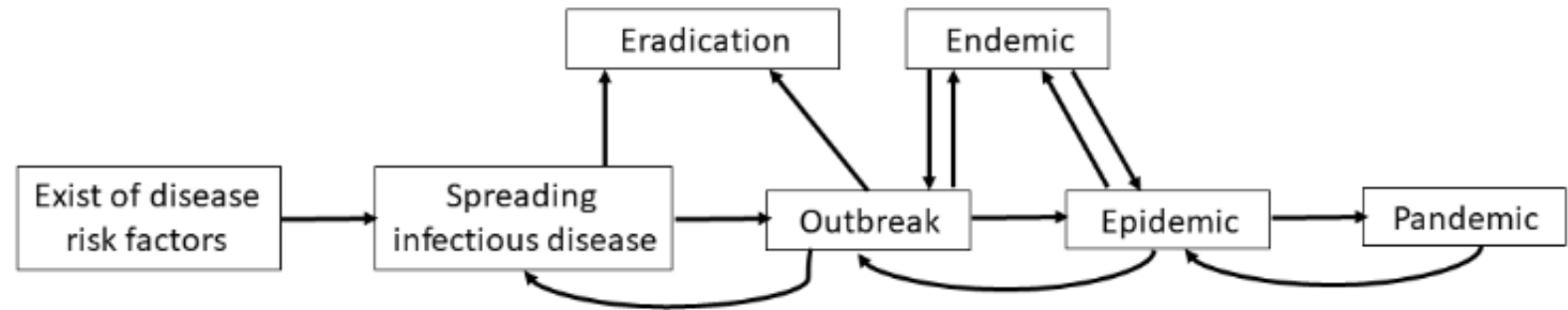
constant maintained increase in occurrences of a disease in a geographic area.

Epidemic

rapid spread of disease to a large number of people in a given population within a short period of time.

Pandemic

spread across a large region, for instance multiple continents, or worldwide.



High likelihood of occurrence annually

Low likelihood of occurrence annually

Low likelihood of severe health and economic impact

High likelihood of severe health and economic impact

Who, when and where

Recall part of the definition on Wikipedia

Epidemiology is the study and analysis of the distribution (who, when, and where)

Who

- **Epidemiology** typically used when dealing with humans, but sometimes also generically when an easy description is sought; e.g., plant disease epidemiology
- **Epizootic**: denoting or relating to a disease that is temporarily prevalent and widespread in an animal population
- **Panzootic** is like a pandemic for animals
- **One Health**: considers health of humans, animals and their environment (including plants)

→ our [who] = humans

Incidence & Prevalence (when?)

Incidence: number of new cases in a population generated within a certain time period

Prevalence: number of cases of a disease at a single time point in a population

→ our [when] = time series

Epidemic curves

- Used to record the occurrence of new cases as a function of time
- When not too many cases, usually "individualised" (bar plots)
- When number of cases is large, continuous curve

→ we will try to have an
overarching continuous curve
as well as
discrete (per node)
timeline / simulation

Some terminology for "where"

- **Epidemic:** diseases that are *visited upon* a population
- **Pandemic:** (will revisit this later in the course) epidemic that has spread across a large region, e.g., multiple continents or worldwide
- **Endemic:** diseases that *reside within* a population
- We don't say "panendemic"

①

The (epidemic) Kermack and McKendrick SIR model

- assumptions
- population remains constant
 - rate of infection $\propto$ Susceptible $\times$ Infected
 - constant rate of people recovering.

The underlying question - What is the size of an epidemic?

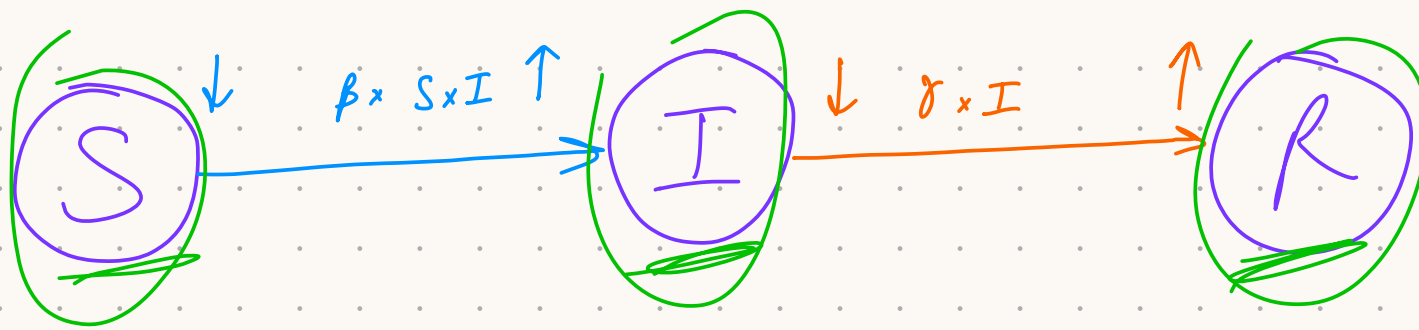
- Suppose we consider the occurrence of an epidemic peak
 - Does it always take place?
 - When it occurs, how bad is it?
 - If an epidemic moves through a population, is everyone infected?

→ mostly "numerical"
(not discrete / per person)

it divides the population in "compartments"
people can only be in one of these ↑ "states"

$SIR \rightarrow$ they are $f(t) =$ no. of people in that state
at time $= t$

- Individuals in the population can be susceptible (S) or infected and infectious with the disease (I). Upon recovery or death, they are *removed* from the infectious compartment (R)



$$\frac{d[S(t)]}{dt} = (-\beta) \cdot S(t) \cdot I(t)$$

$\uparrow$
 constant
 ($\because S$ is decreasing)

contact btw
 S & I people

S, I, R
 are
 unknown $\beta(t)$.

$$\frac{d[I(t)]}{dt} = (+\beta) \cdot S(t) \cdot I(t) - \gamma \cdot I(t)$$

(I is increasing $\because S$
 decreasing $\because R$)

$\uparrow$
 another constant

1600

20%
 30-

$$\frac{d[R(t)]}{dt} = \gamma \cdot I(t)$$

initial conditions At $t=0$ $S=S_0$ $I=I_0$ $R=0$

we know $N = S + I + R$ (obviously)

(total population)

(so, $\frac{dN}{dt} = 0 = \frac{d(S+I+R)}{dt}$)

we do

What is the dynamics of dI/dS ? We have

$$\frac{dI}{dS} = \frac{dI}{dt} \frac{dt}{dS} = \frac{I'}{S'} = \frac{\beta SI - \gamma I}{-\beta SI} = \frac{\gamma}{\beta S} - 1 \quad (3)$$

provided $S \neq 0$

Careful! Remember that S and I are $S(t)$ and $I(t)$ between S and I along solutions to the original ODE

→ thus describes the relationship

$$\frac{\frac{d I(t)}{dt}}{\frac{d S(t)}{dt}} = \frac{+\beta S I - \gamma I}{(-\beta S I)} = \boxed{\frac{\gamma}{\beta S} - 1}$$

$$\frac{dI(t)}{dS(t)} = \frac{\gamma}{\beta S(t)} - 1 \Rightarrow dI(t) = \left(\frac{\gamma}{\beta S(t)} - 1 \right) dS(t)$$

!!!
↓

∫ both

$$I(t) = \frac{\gamma}{\beta} \log[S(t)] - S(t) + K$$

(integration constant)

putting $t = t_0$, we get

$$I_0 = \frac{\gamma}{\beta} \log(S_0) - (S_0) + K$$

$$I = (I_0 + S_0) - \frac{\gamma}{\beta} \log(S_0)$$

put it in

$$I(t) = \frac{\gamma}{\beta} \log[S(t)] - S(t) + (I_0 + S_0) - \frac{\gamma}{\beta} \log(S_0)$$

$$I(t) = I_0 + S_0 - S(t) + \frac{\gamma}{\beta} \frac{\log[S(t)]}{\log[S_0]}$$

now, using $S(t) + I(t) + R(t) = N (\text{constant})$

$$\text{So, } R(t) = N - S(t) - I(t)$$

$$R(t) = N - S(t) - \left(I_0 + S_0 - S(t) + \frac{\gamma}{\beta} \frac{\log[S(t)]}{\log[S_0]} \right)$$

$$= N - \cancel{S(t)} - I_0 - S_0 + \cancel{S(t)} - \frac{\gamma}{\beta} \frac{\log[S(t)]}{\log[S_0]}$$

$$R(t) = \underbrace{(N - I_0 - S_0)}_{= R_0} - \frac{\gamma}{\beta} \frac{\log[S(t)]}{\log[S_0]}$$

$$R(t) = R_0 - \frac{\gamma}{\beta} \frac{\log[S(t)]}{\log[S_0]}$$

$$\frac{\frac{dS(t)}{dt}}{\frac{dR(t)}{dt}} = \frac{(-\beta) \cdot S(t) \cdot \cancel{I(t)}}{\gamma \cdot \cancel{I(t)}} = \boxed{\frac{-\beta S(t)}{\gamma}}$$

$$\frac{dS(t)}{dR(t)} = \frac{-\beta S(t)}{\gamma}$$

$$\frac{dS(t)}{S(t)} = \frac{-\beta}{\gamma} dR(t) \Rightarrow \int \text{both sides}$$

$$\log[S(t)] = \frac{-\beta}{\gamma} R(t) + C$$

C constant

putting $t = t_0$

$$\log[S_0] = \frac{-\beta}{\gamma} R_0 + C$$

$C = \log[S_0]$

putting it in

$$\log [S(t)] = \frac{-\beta R(t)}{\gamma} + \log [S_0]$$

$$\log \left[\frac{S(t)}{S_0} \right] = \frac{-\beta R(t)}{\gamma}$$

Exponent on both sides;

$$S(t) = S_0 \times e^{\left[\frac{-\beta R(t)}{\gamma} \right]}$$

→ we will chart these vals as
graphs,

β = Rate of infection

γ = Rate of recovery of an infected person

Numerically running the SIR model

$$S' = - (10^{-5}) S I$$

$$R' = + (10^{-5}) S I - \frac{1}{14} (I)$$

$$I' = \frac{1}{14} (I)$$

$$\beta = 10^{-5}$$

$$\gamma = \frac{1}{14}$$

at $t = 0$

$$S_0 = 30\,000$$

$$I_0 = 20\,000$$

$$R_0 = 0$$

today

$$S' = -(10^{-5})(3 \times 10^4)(2 \times 10^4) = -6 \times 10^3$$

newly infected

$t=0$

$$I' = +6 \times 10^3 - 1428 = +4571$$

$$R' = 1428$$

recovered

currently infected

people / day.

tomorrow

$$S(1) = S_0 + S'(0) \times 1$$

$$t=1$$

$$= 30\,000 - 6 \times 10^3 = 24\,000$$

$$I(1) = I_0 + I'(0) \times 1$$

$$= 20\,000 + 4571 = 24\,571$$

$$R(1) = R_0 + R'(0) \times 1$$

$$= +1428$$

now calculate S' I' R' again

next day

calculate $\longrightarrow$

S, I, R
use them to get S', I', R' again

Analysis without NOS.

$$S' = \frac{d[S(t)]}{dt} = (-\beta) \cdot S(t) \cdot I(t)$$

$$I' = \frac{d[I(t)]}{dt} = (+\beta) \cdot S(t) \cdot I(t) - \gamma \cdot I(t) = I(t) (\beta S(t) - \gamma)$$

$$R' = \frac{d[R(t)]}{dt} = \gamma \cdot I(t)$$

$$\text{Threshold} = \frac{\gamma}{\beta}$$

Decides how the epidemic will go

Case: $S > \frac{\gamma}{\beta}$

here I' becomes > 0 , infection spreading

Case: $S = \frac{\gamma}{\beta}$

here I' becomes $= 0$, infection staying the same

Case: $S < \frac{\gamma}{\beta}$

here I' becomes < 0 , infection reducing

2

The (endemic) SIS model

assumptions — a fixed no. of people (closed population)

(does have birth & death)

— only two states $\textcircled{S}$, $\textcircled{I}$

— incubation period is $\rightarrow 0$

- $S(t)$ the number of susceptibles at time t
- $I(t)$ the number of infectious/infected at time t
- $N(t) = S(t) + I(t)$ the total population total

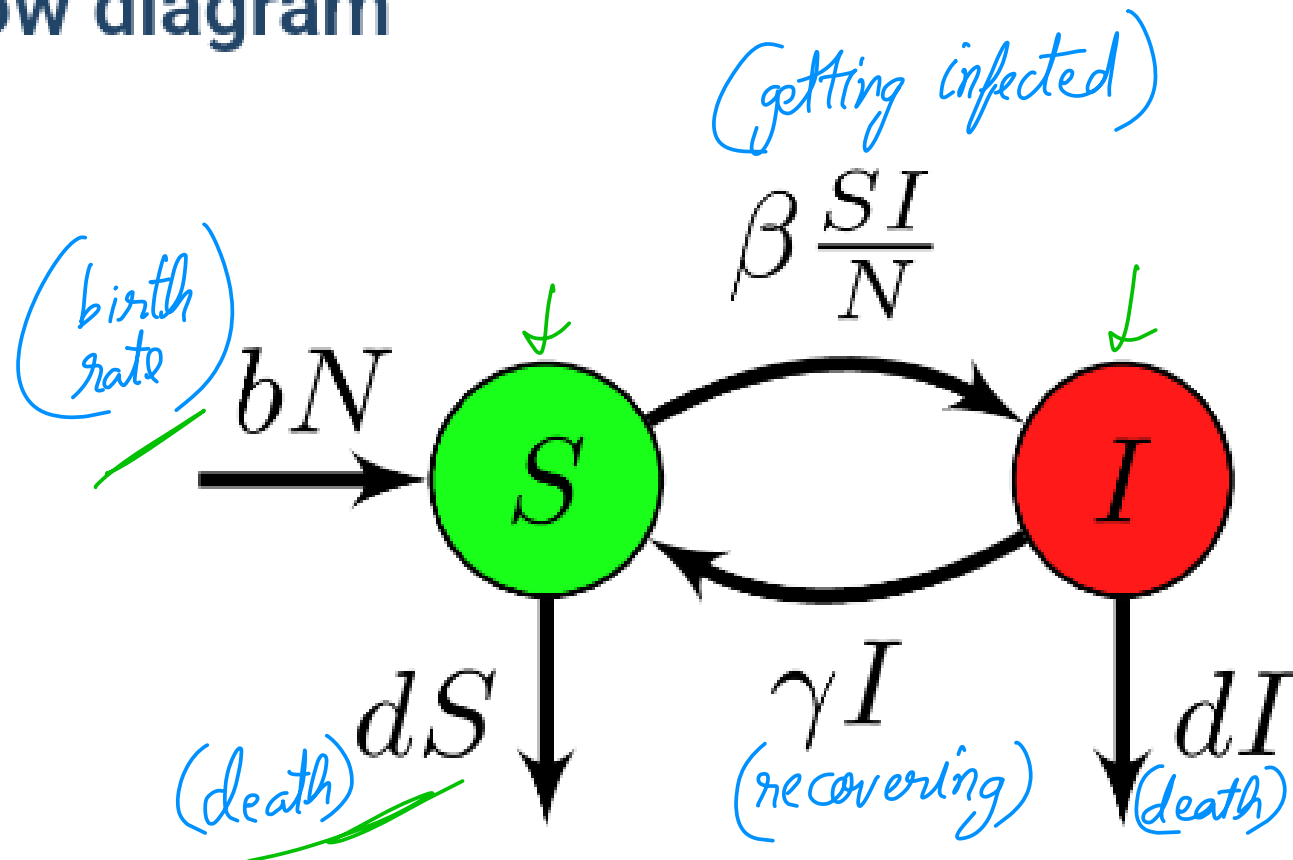
Susceptible individuals

- Are born at a per capita rate b proportional to the total population N
- Die at the per capita rate d , proportional to the susceptible population S
- Newborns are susceptible (vertical transmission is ignored)

Infected and infectious individuals

- Die at the *per capita* rate d , proportional to the infected/infectious population I
- Recover at the *per capita* rate γ
- We do not take into account disease-induced death

Model flow diagram



The model

$$S' = \underbrace{bN}_{\text{birth}} - \underbrace{dS}_{\text{death}} - \underbrace{\beta \frac{SI}{N}}_{\text{infection}} + \underbrace{\gamma I}_{\text{recovery}} \quad (16a)$$

$$I' = \underbrace{\beta \frac{SI}{N}}_{\text{infection}} - \underbrace{dI}_{\text{death}} - \underbrace{\gamma I}_{\text{recovery}} \quad (16b)$$

initial conditions : $S(0) = S_0 \geq 0$
 $I(0) = I_0 \geq 0$

Assume $b = d$ keeps 'N' constant

Dynamics of N

We have

$$\begin{aligned} N' &= (S + I)' \\ &= dN - dS\beta\frac{SI}{N} + \gamma I + \beta\frac{SI}{N} - dI - \gamma I \\ &= dN - d(S + I) \\ &= 0 \end{aligned}$$

As a consequence, for all t , $N(t) \equiv N_0 := S_0 + I_0$

So N remains constant.

Proportions

we use this as a tool;

$$\text{Susceptible people} = s = \frac{S(t)}{N \text{ (constant)}}$$

$$\text{Infected people} = i = \frac{I(t)}{N \text{ (constant)}}$$

Remember

$$s + i = \frac{S(t) + I(t)}{N} = 1$$

now;

$$\frac{d(i)}{d(t)} = \frac{I'}{N}$$

$$= \frac{d I(t)}{d(t)} \times \frac{1}{N}$$

normally

$$\left(\frac{dI}{dt} \cdot N - \frac{dN}{dt} \cdot I \right)$$

N^2

but over N is constant,

∴

$$s = \frac{S}{N} \quad i = \frac{I}{N}$$

Remark that $s + i = (S + I)/N = 1$. The derivative of i is

$$i' = \frac{I'N - IN'}{N^2} = \frac{I'}{N} - \frac{iN'}{N}$$

Since $N' = 0$,

$$i' = \frac{I'}{N}$$

$$i' = \beta \frac{SI}{N^2} - d \frac{I}{N} - \gamma \frac{I}{N} = \beta si - (d + \gamma)i$$

doing $\frac{d(i)}{dt}$

$$I' = \underbrace{\beta \frac{SI}{N}}_{\text{infection}} - \underbrace{dI}_{\text{death}} - \underbrace{\gamma I}_{\text{recovery}}$$

We get

$$\frac{d i}{dt} = \frac{\beta \cdot S \cdot I}{N^2} - d \cdot \frac{I}{N} - \gamma \cdot \frac{I}{N}$$

$$\frac{d(i)}{dt} = \beta \cdot \overset{\substack{\uparrow \\ \text{proportion} \\ \text{of susceptible} \\ \text{people}}}{S} \cdot \overset{\substack{\uparrow \\ \text{proportion} \\ \text{of infected} \\ \text{people}}}{i} - i \left(\overset{\substack{\uparrow \\ \text{death} \\ \text{rate}}}{d} + \overset{\substack{\uparrow \\ \text{recovery} \\ \text{rate}}}{\gamma} \right)$$

Since

$$\delta + i = 1$$

$$\therefore \delta = 1 - i$$

we get

$$\frac{d(i)}{dt} = \beta (1-i) i - i(d+\gamma)$$

↓

$$i' = \beta i - \beta i^2 - i(d+\gamma)$$

$$i' - (\beta - (d+\gamma)) i = -\beta i^2$$

$$i' - (\beta - (d + \gamma))i = -\beta i^2 \quad (18)$$

This is a Bernoulli equation and the change of variables $u = i^{-1}$ gives the linear equation

$$-u' - (\beta - (d + \gamma))u = -\beta$$

so that finally

$$u' + (\beta - (d + \gamma))u = \beta \quad (19)$$



$$\frac{d(i)}{dt} - [\beta - (d + \gamma)]i = -\beta i^2$$

$$u = \frac{1}{i} \quad \frac{du}{dt} = \frac{(-1)}{i^2} \frac{di}{dt}$$

$$\frac{\frac{d(i)}{dt}}{i^2} - [\beta - (d + \gamma)] \frac{i}{i^2} = -\beta$$

$$\frac{1}{i^2} \frac{d(i)}{dt} - [\beta - (d+r)] \frac{1}{i} = -\beta$$

$$-\frac{du}{dt} - [\beta - (d+r)] u = -\beta$$

$$\therefore \frac{du}{dt} + [\beta - (d+r)] u = \beta$$

An integrating factor is

$$\mu(t) = \exp \left(\int P(t) dt \right) = e^{(\beta - (d + \gamma))t}$$

and thus

$$\mu(t)u = \frac{\beta}{\beta - (d + \gamma)} e^{(\beta - (d + \gamma))t} + C$$

for $C \in \mathbb{R}$, so finally

$$u = \frac{\beta}{\beta - (d + \gamma)} + C e^{-(\beta - (d + \gamma))t}$$

now, $u = \frac{1}{i}$ $i = \frac{I(t)}{N_{\text{(constant)}}$

so new initial condition $I(0) = I_0$

The initial condition $i_0 = I_0/N$ takes the form $u(0) = 1/i_0$. Thus,

$$u(0) = \frac{1}{i_0} = \frac{\beta}{\beta - (d + \gamma)} + C$$

which implies that

$$C = \frac{\beta - (d + \gamma) - i_0\beta}{i_0(\beta - (d + \gamma))}$$

get it  (some how ...)

As a consequence, the solution to

$$\begin{aligned} u &= \frac{i_0\beta + (\beta - (d + \gamma) - i_0\beta)e^{-(\beta-(d+\gamma))t}}{i_0(\beta - (d + \gamma))} \\ &= \frac{i_0\beta(1 - e^{-(\beta-(d+\gamma))t}) + (\beta - (d + \gamma))e^{-(\beta-(d+\gamma))t}}{i_0(\beta - (d + \gamma))} \end{aligned}$$

and that of $\dot{i}$ is

$$i(t) = \frac{i_0(\beta - (d + \gamma))}{i_0\beta(1 - e^{-(\beta-(d+\gamma))t}) + (\beta - (d + \gamma))e^{-(\beta-(d+\gamma))t}}$$

In summary, the solution to the system in proportions is given by

$$s(t) = 1 - \frac{i_0(\beta - (d + \gamma))}{i_0\beta(1 - e^{-(\beta-(d+\gamma))t}) + (\beta - (d + \gamma))e^{-(\beta-(d+\gamma))t}}$$

and

$$i(t) = \frac{i_0(\beta - (d + \gamma))}{i_0\beta(1 - e^{-(\beta-(d+\gamma))t}) + (\beta - (d + \gamma))e^{-(\beta-(d+\gamma))t}}$$

The basic reproduction number $\mathcal{R}_0$

Reformulate in epidemiological terms, using the *basic reproduction number*,

$$\mathcal{R}_0 = \frac{\beta}{d + \gamma}$$

We have the following equivalencies

$$\mathcal{R}_0 < 1 \Leftrightarrow \beta - (d + \gamma) < 0$$

$$\mathcal{R}_0 > 1 \Leftrightarrow \beta - (d + \gamma) > 0$$

Also,

$$\frac{\beta - (d + \gamma)}{\beta} = 1 - \frac{1}{\mathcal{R}_0}$$



We have proved the following result

For system (16a)-(16b), the following alternative holds

- If $\mathcal{R}_0 < 1$, then
 - $\lim_{t \rightarrow \infty} s(t) = 1$
 - $\lim_{t \rightarrow \infty} i(t) = 0$, the disease goes **extinct**
- If $\mathcal{R}_0 > 1$, then
 - $\lim_{t \rightarrow \infty} s(t) = 1/\mathcal{R}_0$
 - $\lim_{t \rightarrow \infty} i(t) = 1 - 1/\mathcal{R}_0$, the disease becomes **endemic**

20

Further remarks about $\mathcal{R}_0$

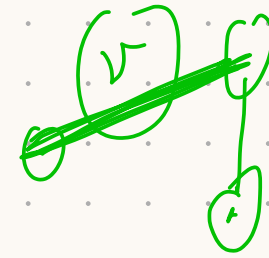
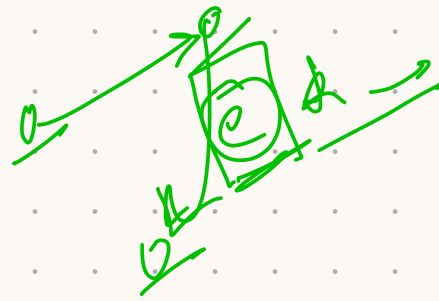
- $\mathcal{R}_0$ determines the propensity of a disease to become established in a population
- The aim of control policies is therefore to reduce $\mathcal{R}_0$ to values less than 1
- The "verbal" definition of $\mathcal{R}_0$ is *the average number of secondary infections produced by the introduction of an infectious individual in a completely naive population*
- Remark that for our basic model, $1/(d + \gamma)$ is the average time of sojourn in the I compartment before death or recovery and β is the probability of infection

The basic reproduction number $\mathcal{R}_0$

Indicator often used in epidemiology. Verbally

average number of new cases generated when an infectious individual is introduced in a completely susceptible population

- If $\mathcal{R}_0 < 1$, then each infectious individual infects on average less than 1 person and the epidemic is quite likely to go extinct
- If $\mathcal{R}_0 > 1$, then each infectious individual infects on average more than 1 person and an epidemic is quite likely to occur



Space for
SIS math...

!!! Want Agent based models !!!

Webinar: Introduction to agent-based modelling for social scientists

1,895 views 2 yr ago ...more



UK DATA SERVICE 5.12K

Subscribe

[Read Each point!]

Table of contents

- Top-down vs. Bottom-up
- Agents and agent-based models
- Simple example
- Complex example
- What to do with the output
- Pros and cons of ABM, platforms/ languages, resources, etc.
- Q&A

agent

5

A problem 'is' top-down if:

- Within a system that is 'whole', 'well-understood' and with central control or structure
- The 'cause' can be isolated to constituent parts, defined interactions, or specific problem areas
- It is well-suited to 'classic scientific method' which focusses on isolation, prediction, repetition, etc.

→ we have a
deterministic
& complete
idea/understanding
of it;

→ A very controlled/isolated/
consistent/well defined prob

#

A problem 'is' bottom-up if:

- Within a poorly-understood, open system with vague parts/interactions and partial/decentralised control
- Not reasonable to assume reducibility, linearity, time/context-independence, or closed/simplicity
- Not easy to apply classic scientific method
- Best understood by trying to recreate it

→ unpredictable,
non-deterministic;

→ very complicated systems, everything is
connected to everything.

→ We decide "based" on the input & expected output on which way we want to go.

Agent based models

Agent-based models

- Multiple
- Interacting
- Rule-based

i.e. $\exists$ "instances" of
"agents" which work together /
interact based on Rules;

Not to Confuse with AI

→ Does everything using its "Own" sensors, actuators etc.

ABM wala Agent → multiple at the same level
→ interaction

What would you use ABM to explore?

Mentimeter

Friendships, collaborations

Complex system dynamics

migration

How policies effect social phenomenon

survival

Agri-Tech and animal farming systems

collective action

conflict

Pest suppression (insects) at the landscape scale by farmers?

Interactions and affects of agents you can't predict the actions of - people, dogs, trams

information dissemination

Riots

Agent-based models have:

A simulated world that has:

- States
- Rules

Recursion!

Objects and/or agents that have:

- States
- Rules

The World

- Can represent any scale (family, city, country, etc.)
- Proceeds through time in discrete 'time steps'
- Includes potentially relevant factors (population, weather, infrastructure, laws, etc.)
- Is unique for each agent as it contains everything *else* (including other agents)
- Has states and follows rules (more on this later)

The Agents

- Can represent most things (people, cars, unis, etc.)
- Includes potentially relevant factors (bank balance, age, colour, speed, competitive ranking, etc.)
- Are unique and behave uniquely
- Have states and follow rules
- Make decisions

States

- Sum of current features
- Can be dynamic/static and can randomness
- Current state can **be influenced by previous state** (of self or others) according to rules
- Current state **can influence future state** (of self or others) according to rules

Rules

- Govern how states change or decisions are made
- Use model generated info (states), modeller choice, randomness
- Can be simple or complex

Example rules:

If [weather = raining] and [temp = cold]
[travel by bus]

Else [travel by bike]

While [interest rates $\leq$ X]
[borrow and spend, do not save]

For each [school-age child in house]
[spend X money on school fees/uniform/etc.]

OK... but why an ABM?

- How things will interact is not intuitive.
- Try to imagine what will happen if we change:
 - Number of agents?
 - Infectiousness?
 - Change of recovery?
 - Average recovery time/duration?
 - Anything else not in simple model (e.g. likelihood of going to hospital, etc.)

What to do with the output?

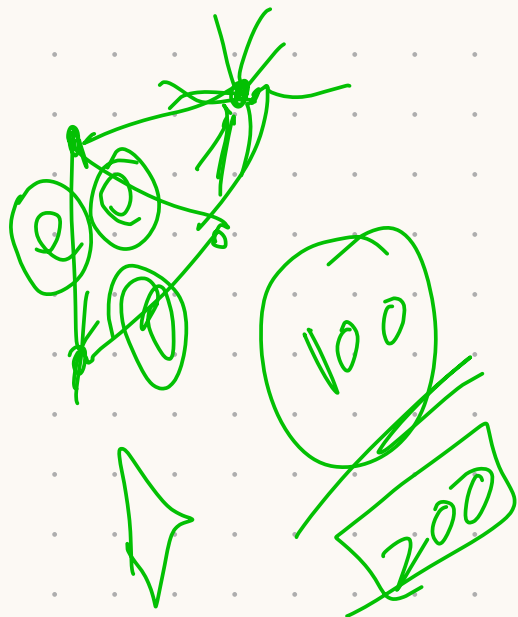
Depends on the ABM, but common options include:

- Parameter sweeps and/or interaction testing
- Outcome probabilities
- Emergent phenomenon identification
- Export to other analysis method (e.g. social network analysis)



Pros of ABM

- No need to rely on intuition for complex behaviours
- Formalise mental models for inspection by others
- Force abstract concepts into concrete representation
- Relatively cheap/easy way to test the 'untestable'
- Potentially very fast
- Can show range of possible futures
- Can use real data, training data, random data, etc.
- Can run multiple times (with or without changes)
- Can be very intuitive



Cons of ABM

- Mental models (and ABM based on them) can be tricky to reflect critically on
- Concrete representations may not adequately capture abstract concepts
- Speed lends itself to 'prediction'
- Possible futures are hard to turn into risk profiles
- Can give false sense of security and truthiness
- Computational methods may motivate distrust/unwarranted trust
- Easy to misunderstand

Summary:

Top-down: reducible, isolatable, straightforward

Bottom-up: non-intuitive large-scale effect of many small acts

Agents: heterogeneous actors that observe, decide, act

ABM: bottom-up simulation with easy testing

Pros: can be inspected, cheap, easy, fast, can do 'impossible' experiments

Cons: simplify complex concepts, misused for prediction, misunderstood

Agent Based Modelling for Social Research – Introduction

4,082 views 2 yr ago ...more



National Centre for Research Methods (NCRM) 21.3K

Methodology on
how to develop ABM's :

5 imp ingredients

ABM design, Programming Language, Data handling,

Model ABM using Cognitive Psychology (For Realism),

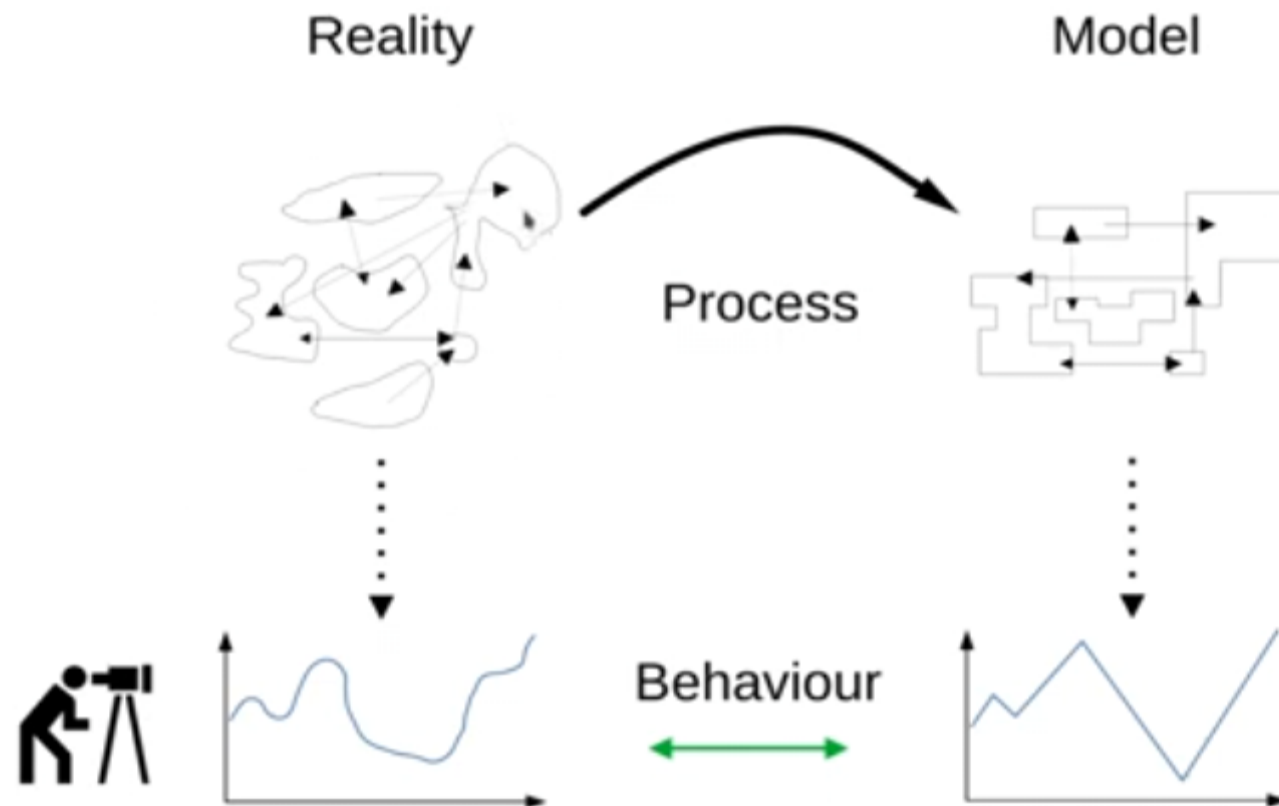
Analytical framework:

Use Bayesian Uncertainty Quantification to analyse &

Calibrate the ABM

A model can NOT answer 'Why' of the event, but it can tell specific outcomes of it based on the initial conditions & parameters we supply from Real World Observations.

What is a model?



Models

- predictive modelling
- explanatory modelling

Explanatory modelling means:

- We observe a phenomenon.
- We have an idea of the conditions that may cause this phenomenon.
- Our intuition is insufficient as a proof.
- Therefore we observe an approximate representation of the putative causing conditions.
- If the representation shows behaviour similar to our observation, our initial idea was correct.

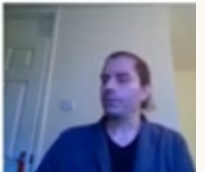
we want to explain / study the process itself, and at the same time get the correct outcomes.

Predictive modelling means:

- We want to predict how a process behaves under given conditions.
- Our intuition is insufficient for a good prediction.
- We do not know enough to describe the process precisely.
- Therefore we observe a formal, approximate representation of the process.
- We use the behaviour of the representation as a prediction of the behaviour of the real process.

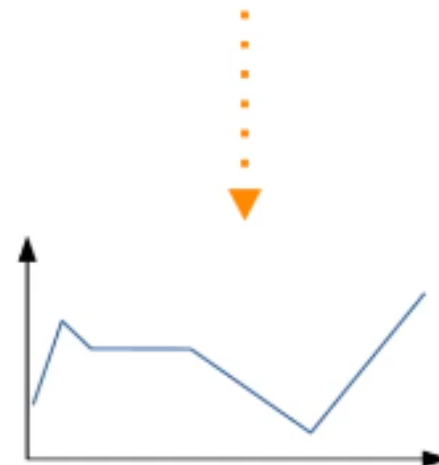
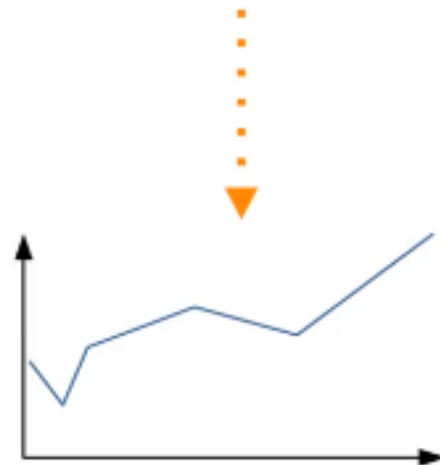
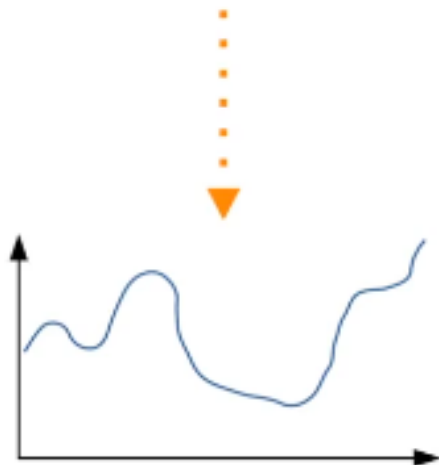
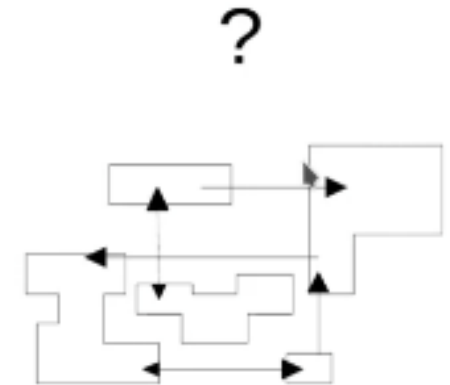
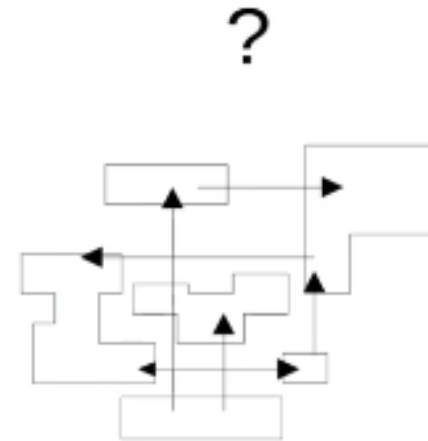
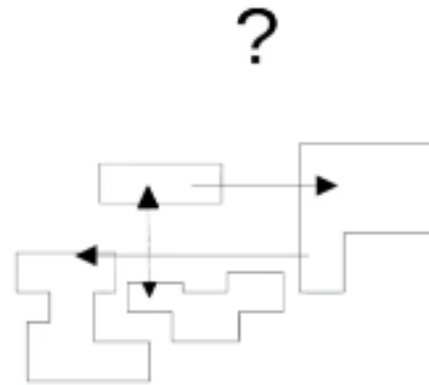
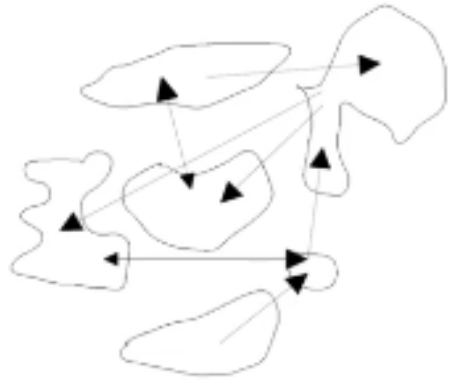
we don't care/look at the process/mechanism, only the outcome;

(our models' outcome matches the observations or NOT?)



explanation

try to build the logic that results in the observed behaviors;



Parting words

- A model is just a tool.
- For a given part of reality (e.g. a river), how we model it depends entirely on what we want to know (e.g. likelihood of flooding, retention of chemicals, carrying capacity of fish populations, etc.).
- A model without a question is a toy.
- A more precise or comprehensive model is not necessarily a better model.
- The only thing a model can show is the consequences of a set of assumptions

*All models are wrong,
but some are useful*

→ George E. P. Box

Complexity

- chaos
- criticality
- emergence

"Emergence"

the phenomenon that complex systems exhibit properties / behaviors iff / due to the interactions btw the individual components (agents in our case).

→ simple components interact
→ due to it complexity increases / properties / behavior of system change
→ irreducible structure "appears" in the system.

"Self Organized Criticality"

"Deterministic Chaos"

Chaos: When the present determines the future but the approximate present does not approximately determine the future.

Small differences in initial conditions, such as those due to errors in measurements or due to rounding errors in [numerical computation](#), can yield widely diverging outcomes for such dynamical systems, rendering long-term prediction of their behavior impossible in general. This can happen even though these systems are [deterministic](#), meaning that their future behavior follows a unique evolution and is fully determined by their initial conditions, with no [random](#) elements involved. In other words, the deterministic nature of these systems does not make them predictable. This behavior is known as **deterministic chaos**, or simply **chaos**. The theory was summarized by [Edward Lorenz](#) as:

- Chaotic systems show huge sensitivity to initial conditions.
- Even simple systems can show chaotic behaviour.
- Many systems in nature actually do.
- Even with exact measurements the behaviour of chaotic systems is unpredictable.

- Even in homogeneous systems consisting of many identical parts, interactions between components can lead to the emergence of structure.
- These interactions can be very simple, but the resulting patterns can be complex.
- Many phenomena in nature are the result of emergence.
- Whether a system shows emergence and what kind of structure will emerge is usually difficult to predict from the behaviour of its components.

the phenomenon where
complex systems evolve towards
a critical state characterized by:

Consider a flat table of finite size with one grain of sand added per unit time interval so that the system has enough time to equilibrate before the next grain drops down. The grains can be added either randomly or at some fixed position of the table. As a result of friction between the grains the system does not automatically equilibrate to a ground state of flat height profile. Initially, the grains are most likely to stay at the same places where they landed, however as we carry on adding more sand, the height profile becomes steeper and small sand

slides or avalanches can occur. If the grain lands on top of other grain it may topple to a lower level overcoming friction due to gravity. This toppling causes local disturbance which does not affect the large-scale picture, i.e. there are no correlations between distant parts of the sandpile. However, as the slope increases, a single grain is more likely to cause other grains to topple and eventually the slope reaches a certain maximal value when the amounts of sand being added and falling off the edges are balanced. Clearly, now the local dynamics no longer governs the process and the avalanches span the entire system leading to complexity. This is the SOC state with its own complex emergent dynamics which cannot be described by local dynamics laws. That is why it is natural to expect that SOC state is robust to the modifications of the systems, which is the crucial requirement for SOC to describe real world. For example, by changing the size of our system (as long as it still stays large), by adding different barriers on the table, by adding some amount of wet sand, the critical state dynamics stays exactly the same.

- small events cascade into larger ones.
- the system becomes very sensitive to external fluctuations.
- the system is highly dynamic & adaptive.
- means that it goes "critical" i.e. somehow the system is holding its state, and even adjusts to local disturbances, BUT the slightest thing can set it off and it undergoes rapid & dynamic changes globally.

- Self-organized criticality means that systems can show spontaneous phase transitions.

- Many systems in nature exist far from equilibrium in a state of criticality.
- It is difficult or impossible to predict phase transitions in systems with self-organized criticality.

Complexity

- Complex systems can show various kinds of surprising, difficult to predict behaviour.
- Deriving simple macroscopic laws of behaviour will therefore often fail.

→ Cannot give a Macroscopic Sol / Law /
Cannot derive the general "formula" of
Complex Systems (in our case ABMs)

how ABM behaves?

results
in

the properties/
behavior of the system.

Interdependence | Non-linearity | Emergence

↓
of agents /
interactions

Cause

Changes in the conditions of
the system leads to wildly
diff. outcomes.

Basic definition of agent-based modelling

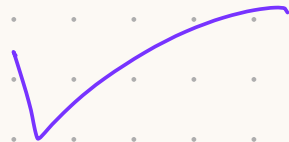
- A type of computer simulation for theory elaboration
- Starts from a social phenomenon and a theory, which is then explicated and implemented in computer code
- The code is submitted to simulation experiments that make it possible to explore the implications of the theory

Discrete/Continuous time Simulation in ABMs;

X bad... has probs.

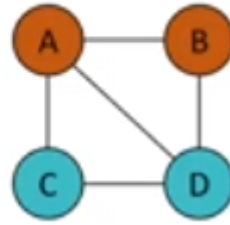


Discrete Event Simulation in ABMs;



Example: A SI Model

- Persons are susceptible or infectious
- They form a network



infectious = A, B
susceptible = C, D

Person

```
| ego.status = "susceptible"
```

```
@ a * ego.neighbors.filter(alter.status = "infectious").size()
```

```
-> ego.status := "infectious"
```

$P(\text{infection}) = a \times (\text{no. of infectious neighbors})$

Person

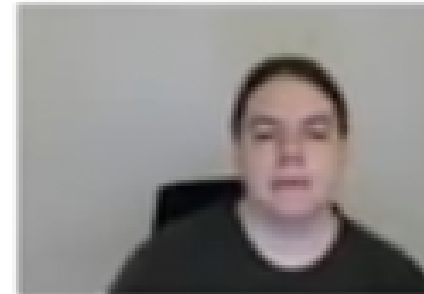
```
| ego.status = "infectious"
```

```
@ b
```

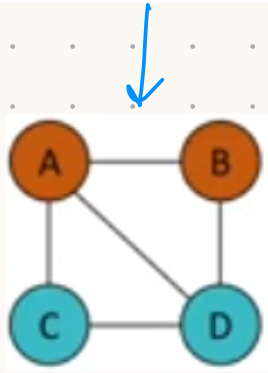
```
-> ego.status := "susceptible"
```

$P(\text{recovery}) = b$

T. Warnke et al. (2017). „Modelling and simulating decision processes of linked lives: An approach based on concurrent processes and stochastic race“
In: Population Studies.

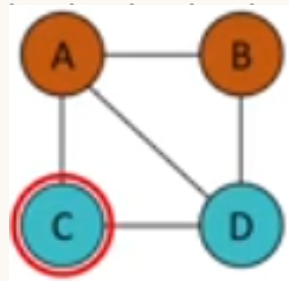


infections
susceptible

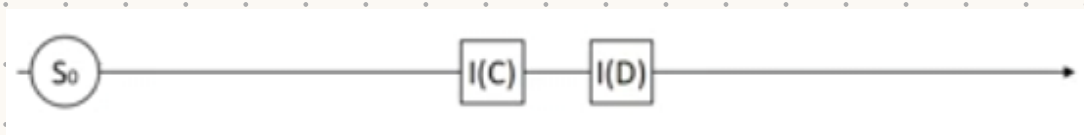
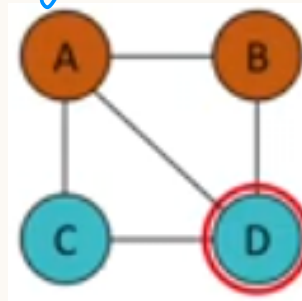


Calculate times for Events

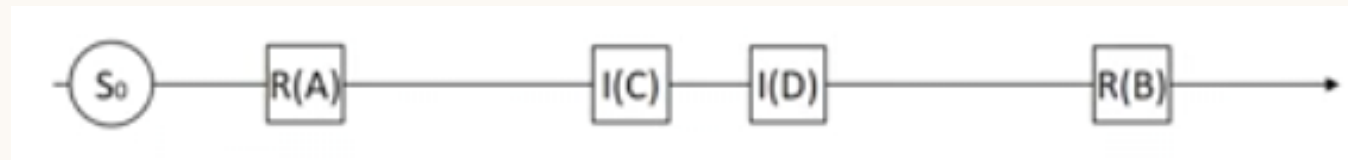
for C



for D

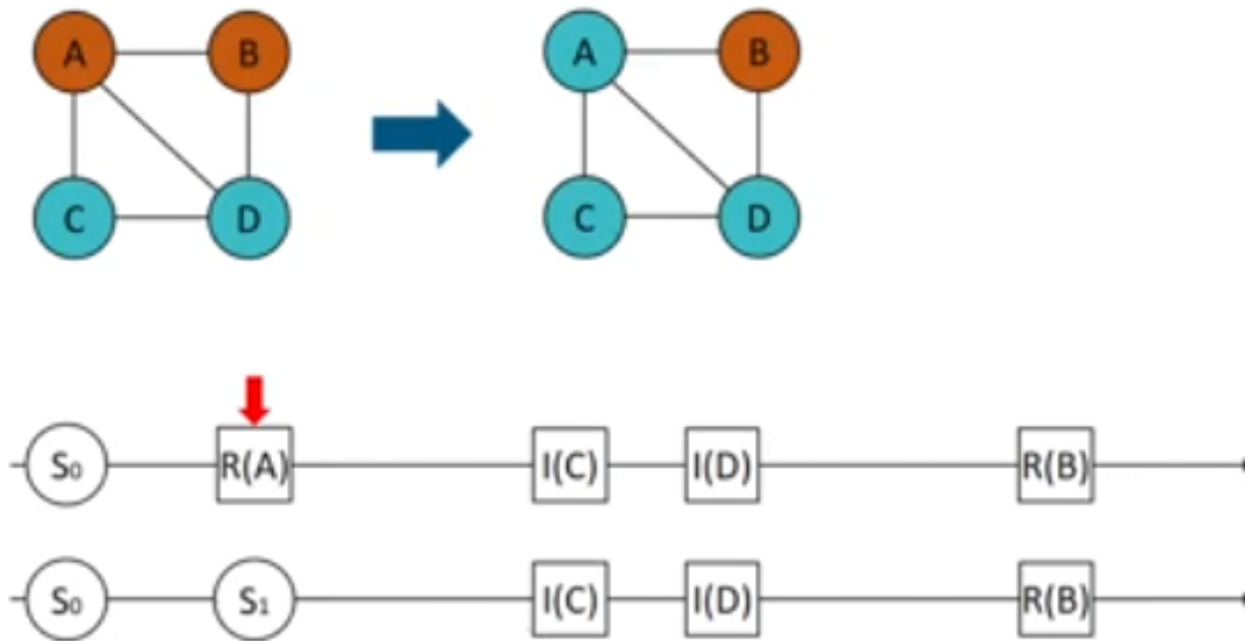


for A for B



After "Scheduling" all events, we execute them
One-by-One & recheck / update the timeline
acc. to how the state of the system changes.

Execute Event



now prob! need to recalculate for other nodes;

Leaving this course here. $\therefore$ it goes into
Hypothesis testing, Statistics etc, we just
want to do simulation. Not test theories.

